

# Additive Structure of Phonological Correspondences

A protoform-agnostic method for discovering mathematical patterns in documented linguistic systems

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## Abstract

Historical linguistics infers ancestral forms and then derives documented ones from them. We invert that order. Representing each attested sound correspondence not as a directed change but as the set of phonological features that differ between two aligned segments — a symmetric *operator* — a language family becomes a repertoire  $O$  of feature-difference vectors over  $\mathbb{F}_2$ , computed with no protoforms, no sound laws, and no externally imposed sound classes. (We say *protoform-agnostic* rather than “reconstruction-free”: cognacy is still detected statistically with LexStat, which is itself a correspondence-based inference — see §3.) We study the geometry and additive structure of  $O$  for Indo-European (IE) and Austronesian (AN), drawn from Lexibank via statistical cognacy (LexStat) and feature-distance alignment (panphon). Four results. (i) Occupancy is not sparsity of the languages but vastness of the space: against  $\approx 1.6\%$  of the algebraic span, the repertoire fills a large fraction of what is actually *available* — 0.29–0.63 (IE) and 0.27–0.48 (AN) depending on whether availability is measured alignment-independently or from the aligner — an order of magnitude denser than the span, not sparse. (ii) The repertoire is additively organized — combining two operators lands back in  $O$  far more often than chance, and this survives a hierarchy of six nested null models up to and including sampling from the corpus’s own opportunity set ( $Z = +6.3$  IE,  $+5.9$  AN;  $p_{MC} \leq 0.004$ ). (iii) That additive structure is a property of the phonological representation, not of genealogy: a concept-permuted control matches it, and branch repertoires overlap no more than random language groupings in IE (whereas in AN they are significantly *more differentiated* than random — a genealogical signal). (iv) The generable-but-unattested operators  $\langle O \rangle \setminus O$  form a concrete geometry of near-misses. The contribution is not a sound law but an *object of study* — the empirical repertoire of correspondences and its measurable additive geometry — together with an inference order that keeps the historical apparatus out of the computation so it can serve as an independent test. A companion manual gives the full didactic treatment.

## 1. Introduction

The comparative method is an inference toward a hidden past: from documented forms  $X$  it reconstructs an ancestor  $\hat{Z}$  and posits the changes  $H$  that produced  $X$ . It answers *how did these forms descend from their common ancestor?* We ask a different question — *treated purely as a set of differences, what shape does the space of attested correspondences have?* — and we answer it in the opposite order:

$$X \longrightarrow \mathcal{M}(X), \quad \text{then, separately,} \quad \mathcal{M}(X) \longleftrightarrow H,$$

where  $\mathcal{M}(X)$  is a mathematical structure discovered without protoforms or precoded laws, and the historical apparatus enters only *afterward*, as contrast. This is not a rejection of reconstruction; it is a more demanding test of it. Protoforms are inferred *from* the documented languages; building the space *on* them and then checking whether it confirms historical regularities risks circularity —

leakage of the target into the inputs. Suspending them during discovery is therefore a design choice, not a limitation.

Two clarifications frame everything below. First, the method is not “theory-free”: transcription, segmentation, concept assignment, the feature matrix, and the aligner are measurement instruments whose influence we make explicit and control for (§3, §4.3). Second, this is emphatically not a claim that “sound change is XOR,” nor a hunt for known sound laws; it is a change of what is measured and in what order.

Two further commitments follow, and we state them plainly for the comparative linguist. First, sound laws are precise about *that* a change occurred and *what* resulted, but silent about *why* one related system selects one outcome where another selects a different one — a “why” that may turn on substrate, contact, and areal pressure, because a language is less a branch off one parent than a new formation confluent from many sources (substrate included as *constitutive*, not noise); such questions live on a network rather than a tree, and become *measurable* only once one can see the whole space of what a system *could* have done (the “holes” of §4.4). Second, and for the same reason, we call two related forms coderivatives rather than *cognates*: the family metaphor (sister, mother, daughter tongues) presupposes descent from a single ancestor, which is exactly the commitment we withhold from the computation. A coderivative is a form co-derived with another, its relatedness evidenced by systematic correspondence; a cognate is the special case of clean descent from one ancestor. None of this denies proto-languages or faults reconstruction — it adds a mathematical dimension alongside the symbolic one, and renames only to keep the question open.

## 1.1 Related work

This work sits inside computational historical linguistics but asks an unusual question of it. Data and tools. We build on Lexibank [1], the aggregated, IPA-standardized wordlist repository; panphon [2] for articulatory feature vectors; LingPy/LexStat [3] for statistical cognate detection; Glottolog [4] for genealogical classification; Concepticon [5] for comparison meanings; and IE-CoR [6] for expert Indo-European cognacy used only in validation (§4.6). Alignment uses Needleman–Wunsch [7]; feature-based phonetic alignment and cognate scoring descend from Kondrak [8] and Dolgopolsky sound classes [9].

Where the field concentrates, and where we differ. Most computational work uses sound correspondences *instrumentally* — as a means to detect cognates, build phylogenies, or date splits: LexStat [3], global-scale phylogenetic inference [10], and cognate-detection benchmarks [11] all treat the correspondence as a step toward a *tree* or an *ancestor*. Typological work such as Blasi et al. [12] tests sound–meaning associations across thousands of languages, again with the correspondence/segment as an instrument. Our object is different: we study the set of feature-difference operators itself — its geometry, additive structure, and occupancy — as the primary phenomenon, and we deliberately withhold the historical apparatus (protoforms, laws, sound classes) from the computation so it can serve as an *independent* test rather than an input. Feature-difference encodings and sound-change typologies exist (e.g. Kümmel’s survey of consonant change [13]); what is new here is not the encoding but treating the resulting *repertoire* as a measurable region of a feature space and characterizing it with explicit null models. Confluence, as support. Our stance that a language is a confluence rather than a branch descends from a long tree-skeptical line — Schmidt’s wave model [17], Schuchardt’s insistence on mixture [18], Trubetzkoy’s Sprachbund [19], and modern phylogenetic networks and linkages [20–22]; we invoke it as *support*, but where those add horizontal edges to a framework of descent (borrowing as deviation from a tree), we withhold descent from discovery and study the geometry of coderivation directly. Mathematics. The additive/geometric apparatus draws on additive combinatorics [14] (sumsets, doubling, additive energy), matroid theory [15] (the dependency structure of the operator set), and coding theory [16] (the span as a linear code, the geometry of unrealized “holes”). To our knowledge these have not previously been applied to phonological correspondence

repertoires.

## 2. The operator and the repertoire

For two aligned segments  $a, b$  with panphon feature map  $\phi$ , the operator is the indicator of differing features

$$\Delta(a, b) = 1[\phi(a) \neq \phi(b)] = \{k : \phi_k(a) \neq \phi_k(b)\} \in \mathbb{F}_2^n.$$

This is a well-defined  $\mathbb{F}_2$  vector for *any* feature alphabet. Over strictly binary features it coincides with  $\phi(a) \oplus \phi(b)$ ; but panphon features are ternary  $(+1, 0, -1)$ , so treating composition as XOR is a *binary approximation* — an identity on binary sub-features whose  $\approx 2\%$  failures (in IE) precisely localize the non-binary dimensions (height), a diagnostic we return to.  $\Delta$  is a *contrast*, not a directed change:  $\Delta(a, b) = \Delta(b, a)$ , so  $\Delta$  does not distinguish  $t \rightarrow s$  from  $s \rightarrow t$ , and “each operator is its own inverse” is a consequence of that symmetry, not a phonological finding. Directed history, when wanted, requires a richer object  $T = (a, b, c, p, \ell_1, \ell_2, w)$  with signature  $\sigma(T) = \Delta(a, b)$ ; this paper studies the symmetric signatures. We keep operators of one to three features (atoms and small molecules).

The forms compared are coderivatives (§1): related by a shared, evidenced history — recurrent correspondence — without presupposing a single reconstructed ancestor. A family’s repertoire  $O$  is its set of recurrent operators over those coderivatives.  $O$  generates a subspace  $\langle O \rangle = \text{span}_{\mathbb{F}_2}(O)$  of dimension (rank)  $r$ ; crucially  $O$  is not a subspace (not closed under XOR, does not contain 0) — it is a *region*, and the object of interest is the relation between the region  $O$  and the capacity  $\langle O \rangle$ .

## 3. Data and method

Corpus. Lexibank word lists, IPA-segmented, for Indo-European (304 languages available) and Austronesian (978), taking per family the languages with the most forms. Cognacy is detected statistically with LexStat (LingPy); alignment is Needleman–Wunsch with substitution cost equal to the fraction of panphon features that differ; features are a 12-feature primary subset of panphon — `cont`, `voi`, `nas`, `ant`, `cor`, `lab`, `back`, `round`, `strid`, `hi`, `lo`, `son` (manner, place, laryngeal, and major-class features; panphon’s full set is larger, and the subselection is a documented, auditable choice, swept in the sensitivity analysis). No protoforms, laws, or sound classes enter the computation. Operators are taken from aligned, non-identical, consonantal positions with support thresholds (30 at family scale); vowel correspondences are set aside in this pilot (they are dominated by ablaut/reduction in IE and would need a graded treatment — a deliberate scope limit, not a claim that they are noiseless). Independent validation uses IE-CoR expert cognacy (§4.6). Every quantity is reproducible from a single make target (§7).

On “protoform-agnostic.” We withhold protoforms, sound laws, and sound classes from the discovery computation. We do *not* claim to be free of all historical inference: statistical cognacy (LexStat) learns language-pair sound correspondences and clusters by them, so a genealogical/correspondence assumption is present in how cognate sets are formed. But that assumption is the Neogrammarian one — cognacy evidenced by *recurrent, systematic correspondence* (what LexStat measures), which is the evidence *for* cognation, not the reconstructed ancestor inferred on top of it. We keep this foundation and set aside only the reconstructive step. Filtering the input to reconstruction-certified cognates would defeat the design — it would leak the historical answer into the data and forfeit the independent test — so what is kept out is the *reconstructed ancestor and the pre-stated laws*; §4.6 quantifies how much LexStat’s choices shape the repertoire, and §4.3 controls for representation. The honest scope is therefore “protoform- and sound-law-free,” not “assumption-free.”

Two families give a minimal comparative frame; “IE” and “AN” are pilot genealogical boundaries, not results.

## 4. Results

### 4.1 Occupancy: sparsity is mostly algebraic

IE uses  $|O| = 67$  operators, AN  $|O| = 22$ , with ranks  $r = 12$  and  $r = 10$ . Against the full algebraic span these are 1.6% and 2.2% — but the span  $2^r - 1$  contains feature-bundles that no pair of real segments could realize. Nested between  $O$  and  $\langle O \rangle$  are the realizable universe  $U_S = \{\Delta(a, b) : a, b \in S\}$  (over the attested segment inventory  $S$ ) and the opportunity universe  $\Omega_D$  (differences that actually occur in aligned corpus positions):

| family        | $\rho_{\text{alg}}$ (span) | $\rho_{\text{seg}}$ (realizable $U_S$ ) | $\rho_{\text{opp}}$ (aligner $\Omega_D$ ) | $\rho_{\text{ind}}$ (alignment-free) |
|---------------|----------------------------|-----------------------------------------|-------------------------------------------|--------------------------------------|
| Indo-European | 0.016                      | 0.48                                    | 0.63                                      | 0.29                                 |
| Austronesian  | 0.022                      | 0.43                                    | 0.48                                      | 0.27                                 |

where  $\rho_{\text{alg}} = |O|/(2^r - 1)$ ,  $\rho_{\text{seg}} = |O|/|U_S \cap \langle O \rangle|$ , and  $\rho_{\text{opp}} = |O|/|\Omega_D \cap \langle O \rangle|$ . Because  $\Omega_D$  is built from our *own* alignments,  $\rho_{\text{opp}}$  could be circular; we therefore add an alignment-free opportunity  $\Omega_{\text{bag}}$  (differences between consonants co-present in same-concept words, no alignment used) giving  $\rho_{\text{ind}}$ . The aligner did inflate the figure —  $\rho_{\text{ind}}$  is roughly half of  $\rho_{\text{opp}}$  — but the conclusion holds under a denominator that never touches our aligner: at 0.27–0.29 (and  $\rho_{\text{seg}} \approx 0.43$ –0.48, also alignment-free) the repertoire is an order of magnitude denser than the span. The “98% unused” is the algebraic space being enormous, not languages being restrictive.

### 4.2 The repertoire is additively organized

Composing two operators is XOR of their feature-sets. The composition-realization index

$$C(O) = \frac{|\{\{u, v\} \subseteq O : u \oplus v \in O\}|}{\binom{|O|}{2}}$$

is 0.220 (IE) and 0.234 (AN); conditioned on the opportunity universe,  $C_{\Omega}(O) = 0.81$  (IE), 0.66 (AN), and conditioned on the alignment-free opportunity  $\Omega_{\text{bag}}$  (§4.1) it remains substantial at 0.57 (IE), 0.48 (AN) — the composition-closure is not an aligner artifact. To test whether this exceeds chance we compare against a hierarchy of null repertoires, each preserving more structure and each sampled to size  $|O|$ ; we report  $Z$  and empirical  $p_{\text{MC}}$  over 500 simulations:

| null preserves                  | IE $Z$ | AN $Z$ |
|---------------------------------|--------|--------|
| size + weight $\leq 3$          | +15.7  | +12.0  |
| Hamming weights                 | +14.8  | +9.0   |
| exact margins (swap-MCMC)       | +7.5   | +3.6   |
| rank / span $\langle O \rangle$ | +45.6  | +13.7  |
| realizable $U_S$                | +9.2   | +7.3   |
| opportunity $\Omega_D$          | +6.3   | +5.9   |

All six survive at  $p_{\text{MC}} \leq 0.004$  (500 simulations, the resolution floor; a larger  $B$  and FDR control across the null  $\times$  family  $\times$  statistic grid are scheduled — see the research programme). We describe the six as “increasingly constrained” rather than a strict nesting: the span null (row 3) samples a different ambient set than the others, so the  $Z$  column is not monotone. Even drawn from exactly the differences the corpus made available, the observed repertoire is significantly more composition-closed. Three further additive statistics agree (against the  $\Omega_D$  null), each on unordered tuples to match  $C(O)$ : triple density  $\tau(O) = |\{\{u, v, w\} \subseteq O : u \oplus v = w\}|/\binom{|O|}{3}$  is high (IE  $Z$ =+6.6, AN +6.5);

the doubling constant  $\kappa = |O \oplus O|/|O|$  (with  $O \oplus O = \{u \oplus v\}$ ) is low (IE  $Z=-5.1$ , AN  $-6.0$ ; small doubling = strong additive structure); and the additive energy  $E(O) = |\{(a, b, c, d) \in O^4 : a \oplus b = c \oplus d\}|/|O|^3$  is high (IE  $+7.4$ , AN  $+7.3$ ). The repertoire behaves like an approximate (possibly affine) subspace. Its dependency structure has a basis-independent core: the vector matroid of  $O$  (operators as  $\mathbb{F}_2$  vectors) has 162 (IE) and 18 (AN) three-element circuits (minimal XOR-zero triples), with  $\{\text{voi}\}$  (voicing) in the most of them — the additive hub. (*Circuit and hole counts scale with  $|O|$ , so IE-vs-AN raw counts are not directly comparable; the size-controlled comparison is the  $Z$  against nulls.*)

### 4.3 What the additive structure is — and is not

The structure of §4.2 is a property of the phonological representation and inventory, not of genealogy or semantics. Two controls establish this. First, permuting concept labels to destroy all cognacy and semantic linkage (regime  $D_R$ ) yields additive structure *as high as or higher than* expert cognacy  $D_G$  ( $C=0.215$  vs  $0.169$ ), so the type-level additive closure does not distinguish history from inventory. Second, the one linear dependency we observe — in AN,  $\text{ant} = \text{cor} \oplus \text{lab}$  — is a property of the *selected* repertoire, not of the feature matrix: the realizable universe  $U_S$  has full rank (12, corank 0), so the ontology would permit independence; the dependency appears only in  $O_{\text{AN}}$ . The additive geometry is real (§4.2) but must be read as *representational*; genealogical signal lives elsewhere (§4.5–4.6).

### 4.4 The geometry of holes $\langle O \rangle \setminus O$

Among the operator-shaped (weight  $\leq 3$ ) vectors the span generates, most are holes — generable but unattested. For each hole  $x$ , its depth is  $d(x, O) = \min_{o \in O} d_H(x, o)$ .

|               | generable (weight $\leq 3$ ) | observed | holes | at $d = 1$ | holes in $\Omega_D$ |
|---------------|------------------------------|----------|-------|------------|---------------------|
| Indo-European | 298                          | 67       | 231   | 160        | 40                  |
| Austronesian  | 119                          | 22       | 97    | 53         | 24                  |

Most holes lie one feature from an attested operator; the linguistically loaded ones are the holes in  $\Omega_D$  (40 IE, 24 AN) — differences the corpus made available yet the system left unused. These are the concrete candidates for interpretation: absences that are choices, not impossibilities.

### 4.5 Systems and branches

Partitioning each family into Glottolog branches and taking  $O_F = \bigcup_r O_r$ , the persistence histogram  $H_k$  (operators in exactly  $k$  branches) decomposes the repertoire without double-counting: IE  $|O_F| = 97$ , strict nucleus  $K_{\cap} = 12$  (the atoms  $\{\text{ant}\}$   $\{\text{cont}\}$   $\{\text{voi}\}$  in every branch), majority nucleus  $K_{1/2} = 38$ ; AN  $|O_F| = 48$ ,  $K_{\cap} = 4$ ,  $K_{1/2} = 18$ . But raw branch overlap is uninterpretable without a null. Against a grouping null (arbitrary same-size language groups) the mean Jaccard tells opposite stories in the two families:

|               | mean Jaccard (real branches) | arbitrary groupings | $Z$                                        |
|---------------|------------------------------|---------------------|--------------------------------------------|
| Indo-European | 0.42                         | $0.45 \pm 0.05$     | $-0.7$ (indistinguishable)                 |
| Austronesian  | 0.51                         | $0.57 \pm 0.01$     | $-5.4$ (real branches more differentiated) |

At the level of operator types, Indo-European branches share no more than random groupings — the apparent “shared core” is shared *inventory* — whereas Austronesian branches are significantly more

differentiated than chance. Under rarefaction the nucleus size is sample-sensitive, so branch *richness* is not interpreted; the robust facts are the histogram decomposition and the grouping-null contrast.

## 4.6 Validation against expert cognacy

Statistical cognacy could contaminate the repertoire with false cognates. Against IE-CoR expert cognacy the contamination is real at the instance level (pair precision 0.73, recall 0.33) but absorbed at the type level: LexStat’s operator inventory has precision  $\approx 1.0$  and recall 0.66 against the expert repertoire — a false cognate almost always produces an operator already present. The type inventory is thus more robust than individual cognate identification. Where genealogical signal does live is the distributions: mutual information between operator and branch is modest but non-zero and larger in AN than IE (0.22 vs 0.14 normalized), consistent with §4.5; a few operators dominate each family (effective count  $N_{\text{eff}} = 24$  of 119 IE, 18 of 63 AN — where 119/63 are the *instance-level* type inventories before the support threshold that defines the repertoire  $O$  of 67/22, distinct again from the branch-union  $O_F$  of 97/48 in §4.5).

## 4.7 Sensitivity: the headlines are measurement-invariant

The paper flags transcription, segmentation, the feature matrix, and the thresholds as *measurement instruments* (§1, §3); a sensitivity sweep confirms the headline signs do not depend on them. Sweeping the support threshold (15–60), IE’s rank stays 12 and  $C(O)$  stays 0.215–0.224 (AN: rank 10–12,  $C(O)$  0.17–0.23); sweeping the operator weight cap ( $\leq 2, 3, 4$ ) and the feature subset (the full 12, minus lab, minus {round, lo}, and a manner+voice+place 8-set) leaves  $C(O)$  in 0.21–0.37 (IE) and 0.14–0.28 (AN) — in every case well above the size-matched null floor ( $\approx 0.12$ –0.15, §4.2). Occupancy  $\rho_{\text{opp}}$  stays 0.5–0.7 (IE) and  $\sim 0.4$ –0.5 (AN) across thresholds. The one soft spot, reported honestly, is Austronesian at weight cap  $\leq 2$  ( $C(O) = 0.14$ ,  $|O| = 12$ ), where the repertoire is simply too small to resolve. Reproducible: make sensitivity FAMILY="X". (The LexStat-threshold and aligner sweeps, which require the corpora, are scheduled.)

## 5. Figures

Two graph types make the repertoire visible; each figure below carries a self-contained caption. Figures 1–2 are the *segmental correspondence graph*  $G_S$  for Indo-European and Austronesian: nodes are segments, edges are recurrent correspondences coloured by change type. Figure 3 is the *operator graph*  $G_O$  for Indo-European: nodes are operators and an edge joins  $u, v$  whenever their composition  $u \oplus v$  is itself an operator of the repertoire — so  $G_O$  is the visual face of the composition index  $C(O)$  of §4.2. (The two graphs, together with a *language graph*  $G_L$  over operator distributions, are the three-graph architecture the companion manual develops.)

## 6. Discussion

The contribution is an object and an order. The object is the empirical repertoire  $O$  of symmetric feature-difference operators, and the separation between it and the subspace  $\langle O \rangle$  it generates: a system uses a small, additively-structured region of a large algebraic space, densely relative to opportunity, with a concrete geometry of unused near-misses. The order is discover-then-contrast, which keeps reconstructions, laws, and sound classes out of the measurement so each can serve as an independent yardstick later.

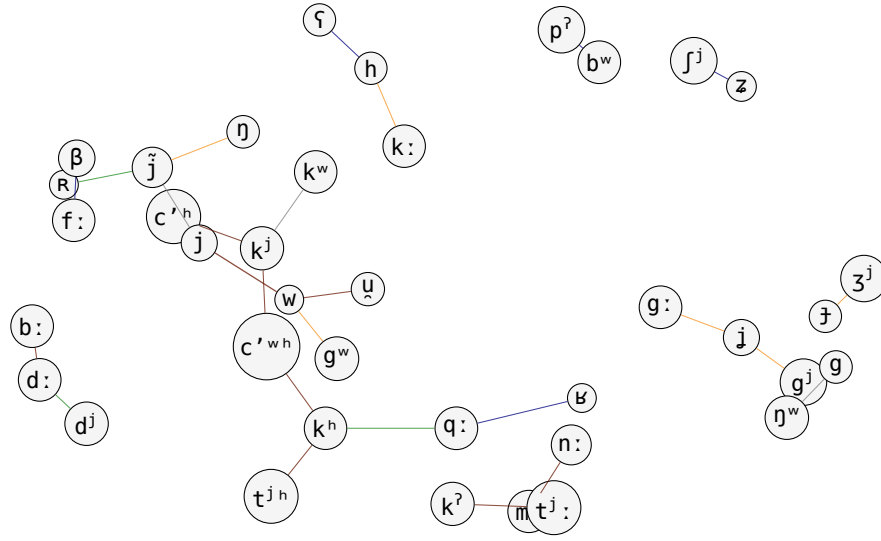


Figure 1: Segmental correspondence graph  $G_S$  of Indo-European. Nodes are segments; edges are recurrent correspondences, with thickness  $\propto$  frequency and colour by change type: **voicing**, **fricativization/lenition**, **palatalization/height**, **place**, **prenasalization**.

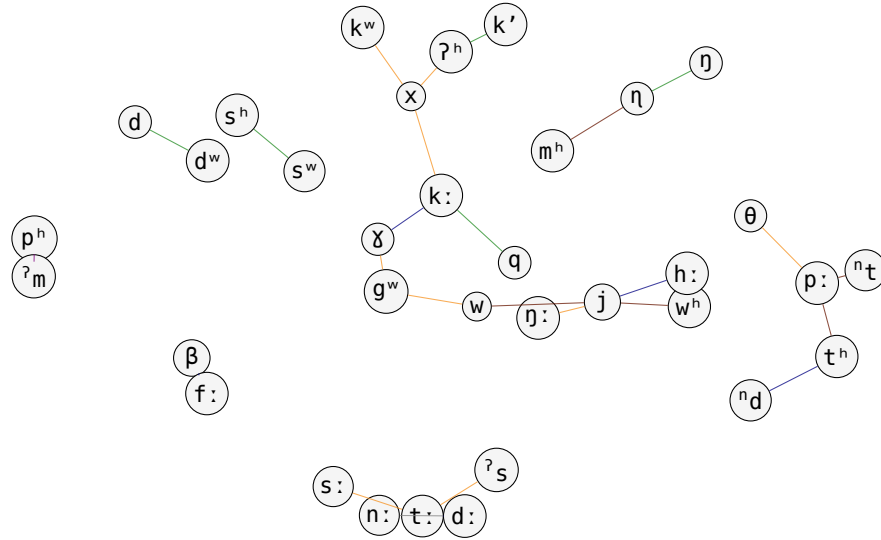


Figure 2: Segmental correspondence graph  $G_S$  of Austronesian. Nodes are segments; edges are recurrent correspondences, with thickness  $\propto$  frequency and colour by change type: **voicing**, **fricativization/lenition**, **palatalization/height**, **place**, **prenasalization**.

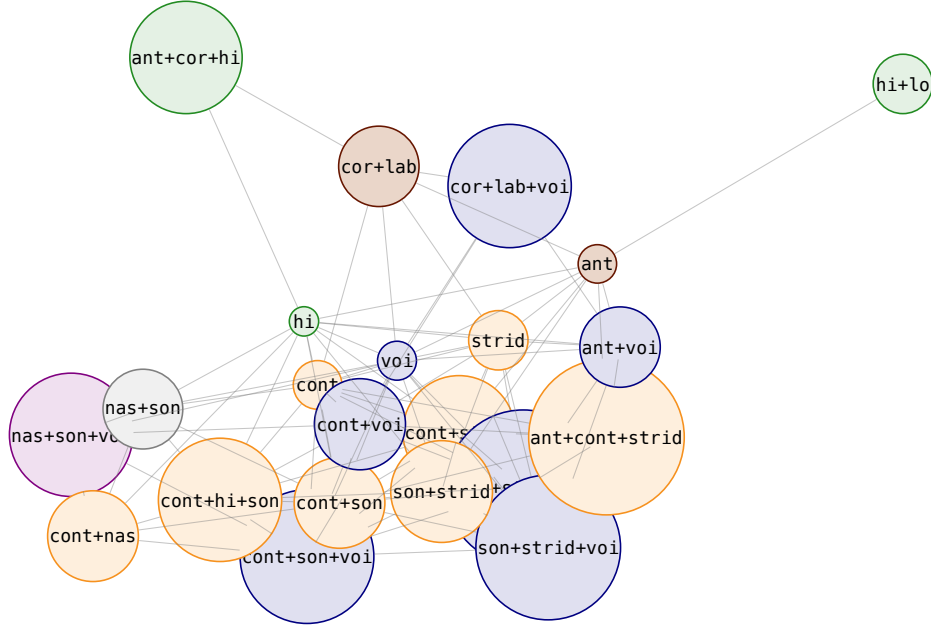


Figure 3: Operator graph  $G_O$  of Indo-European. Nodes are operators (the 22 with highest support), sized by frequency and coloured by change type (voicing, fricativization, palatalization/height, place, prenasalization). An edge joins  $u, v$  when their composition  $u \oplus v$  is itself an operator of the repertoire — so this is the graph of *additive closure*. Dense connectivity is the visual face of the composition index  $C(O)$ .

Two boundaries keep the claims honest. The additive structure is *representational*, not genealogical (§4.3); the genealogical question is properly posed on distributions and, in future work, on a directed, context-conditioned layer — the tensor  $X_{ijoc}$  of operator frequency by language pair and context, whose non-negative factorization and minimum-description-length rules would let sound laws *emerge* as compressive bundles rather than be inserted. Under a historical-holdout protocol (discover; test stability; cluster languages blind; only then compare to reconstructions and trees), a recovered law would be evidence precisely because it was never an input. The same discipline — discover structure first, admit ground-truth labels only to test — is a template for leakage-free evaluation of language models on historical and typological tasks.

## 7. Limitations and reproducibility

**Limitations.** Two families are a pilot; “family” is a genealogical label taken as a boundary, not a result. The opportunity universe  $\Omega_D$  and the additive nulls depend on the corpus and aligner. The additive structure is type-level and representational; loans-as-system-influence could not be tested (loan annotations are absent in the pilot corpora and require a resource such as WOLD). Metathesis — a *reordering* of segments, as in the classic  $TR \leftrightarrow RT$  — is not captured: the alignment is monotonic (order-preserving), so a metathesized pair does not yield a “metathesis operator” but surfaces as spurious substitutions and gaps (largely filtered by the alignment-cost cohesion screen, but a genuine gap). Metathesis is properly a *permutation* of skeleton positions, an operator on order rather than on the features of a slot — an orthogonal dimension (the symmetric group  $S_k$  beside the feature space  $\mathbb{F}_2^n$ ) that a future model should add explicitly (see the research programme). Force-directed figures lack an articulatory coordinate system. None of these affect the null-controlled results, but they bound their interpretation.

**Reproducibility.** All results run from transformations/src against a repertoire built by make



family FAMILY="X". Key targets: make universes (§4.1), make nulls / make additive (§4.2, §4.4), make repr-control (§4.3), make superposition (§4.5), make cognate-eval / make distributions (§4.6). Datasets: Lexibank, IE-CoR, Glottolog (cached classification). LexStat threshold 0.55, 100-run scorer; panphon 12-feature primary set; seeds fixed. The companion manual (docs/manual) gives worked, hand-computed examples of every measure and the full derivations.

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